Requests Responses and JSON

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

Requests, Responses, and JSON

> Prepare Requests Responses and JSON: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Parse **JSON** into Python records

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min

> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Retrieval answer guidance

Tools and APIs

Without opening the old lesson, explain one key idea from Tools and APIs. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of tools and api **contract**s

Reference example, not the only acceptable wording: Teacher explanation: Tools expose structured operations with **input** and **output contract**s.

Generating a tool call is a proposal; validation and authorisation decide whether it executes. For lookup_room accept exactly room_id as a nonempty string; return found plus capacity or a not-found result. It is read-only.

Reject unknown arguments and do not replace missing values with invented facts.

User Need and AI Contribution

Without opening the old lesson, explain one key idea from User Need and AI Contribution. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of capstone requirements and corpus

Reference example, not the only acceptable wording: Teacher explanation: A grounded-assistant capstone begins with bounded requirements and a versioned permitted corpus.

A baseline can be a manual lookup **process** using the same **evidence**. Specify the user question, source authority, access, **output**/fallback **contract** and evaluation plan. Freeze corpus IDs and split cases before implementation.

Keep teacher material out of student retrieval.

Verify Numbers and Dates

Without opening the old lesson, explain one key idea from Verify Numbers and Dates. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of claim support and citation evaluation

Reference example, not the only acceptable wording: Teacher explanation: Citation evaluation checks source existence, exact passage support and coverage of answer claims. A real citation can still be irrelevant.

For this classroom rubric count a claim as supported only when its cited passage entails it. Separately count correct citation links. Use the number of factual claims as the support denominator.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Request id=7 returns **status**=200 and {"id":8,"pages":90}. Explain why a successful **status** is insufficient and specify the handling.

Concept to explain

Week 22: **Request**s, **Response**s, and **JSON**

Term	Meaning
request	A message asking a system to return information or perform an action.
response	The information or status returned after a request.
JSON object	A collection of named fields and values written using JSON syntax, such as {"pages":80}.
status	A reported condition of a request or process, such as success, pending or error.

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Alternative compact allocation

> This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Teach Week 26 as a two-session investigation using fictional data. A fictional activity service supplies **JSON** text. The **input contract** is a **JSON object** with name text and minutes an integer from 0 through 60.

The **output contract** will be a record with accepted and minutes fields. **Parsing** valid **JSON** is only the first step. Assess the three stated outcomes through the independent task, not by copying the worked result.

Readiness answer

Reading obtains characters; validation checks whether their meaning and values satisfy rules.

Python 3 editor with a fresh local practice file

Paper trace table and the student workbook

Fictional **input** records from this lesson; no real student identifiers

End goal for the pictorial case

By the end, I can explain **request**s **response**s and **json** using a traced example and a stated limitation.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

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Explain the mechanism

> Model Requests Responses and JSON using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

1. **Request** requires question:text and source_ids:list.

2. **Response** requires answer:text, citations:list, **status**:text.

3. Missing citations or wrong source_ids type fails validation.

4. Error **response** should state code, message, and safe next step.

Recalculate, retrace, or restate the reasoning in your own representation:

> CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

Concrete teaching case

> ### The teaching case
>
> **Request** is for B7; **response** is {"id":"B7","pages":120}. Another **response** has malformed **JSON**.

> What checks precede use?

Exact explanation to model

> ### Why the result follows
>
> **Request** success, **JSON** validity, matching id, required fields and numeric type. Handle malformed **response**s explicitly.

Explain the added worked example

A fictional activity service supplies **JSON** text. The **input contract** is a **JSON object** with name text and minutes an integer from 0 through 60. The **output contract** will be a record with accepted and minutes fields.

Parsing valid **JSON** is only the first step.

Explain the added worked example · 1

The **output** is {"accepted": true, "minutes": 12}. **JSON** uses lowercase true, false and null; Python uses True, False and None. **json**.loads converts between representations.

The worked **output** line assumes the parsed top-level value is a dictionary; a robust **interface** must reject other top-level types before accessing fields.

Key terms

Term	Meaning
JSON	A text format for structured values
Parsing	Converting formatted text into program values
Interface	The agreed way one part of a system exchanges data with another
Contract	Specified inputs, outputs and error behavior

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

> **JSON text contains {"name": "Sketch", "minutes": "12"}. Does parsing fail, or does validation fail?**

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Requests Responses and JSON

A lunch-count **response** must satisfy a declared **JSON contract**.

The facts we will use

> **Supplied facts**
>
> The toy **response** has exactly class_code text and portions integer 0..30. Example: {"class_code":"R8","portions": 12}.

> **Supplied facts**
>
> All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-3/22

Requests Responses and JSON

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Read fields

class_code is nonempty text; portions is the integer 12.



2. Check range

0 <= 12 <= 30, so the declared count is within range.



3. Accept structure

Accept this response shape; acceptance does not prove that 12 portions were actually counted.

CHANGE ONE THING

Replace 12 with the text "12".

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Read fields	class_code is nonempty text; portions is the integer 12.
2. Check range	0 <= 12 <= 30, so the declared count is within range.
3. Accept structure	Accept this response shape; acceptance does not prove that 12 portions were actually counted.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Read the picture in order. Accept this **response** shape; acceptance does not prove that 12 portions were actually counted. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

Keep the stated limit

JSON well-formedness, schema validity and real-world truth are different checks.

Change one thing in the pictured case

Change one thing

Change: Replace 12 with the text "12".

Predicted result

Prediction: Reject the **response** under this strict **contract**.

Why the result follows

Reason: Digits inside quotes represent text, whereas portions requires an integer.

Ask while showing the picture

Ask while showing the picture · 2

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate Requests Responses and JSON, reduce support deliberately and check individual reasoning before discussing answers.

Board sequence

1. Put the facts on the board

Write the given facts: **Request** is for B7; **response** is {"id":"B7","pages":120}. Another **response** has malformed **JSON**.

2. State the decision to explain

Write the question to explain: What checks precede use?

3. Reveal the worked reasoning

Reveal after **prediction**s: **Request** success, **JSON** validity, matching id, required fields and numeric type. Handle malformed **response**s explicitly.

Hold back the independent task

Keep for an independent attempt: An API returns {"ok":false,"error":"not_found"}. Your app tries to display **response**.title. What should it do instead?

Classroom dialogue

Ask the class

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Language to model

Say: Predict first. Show the rule, information path or calculation that would produce the

> result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

Parsing succeeds; validation fails because minutes is text. Exit: No. Syntax, schema validity and **observation**al truth are separate checks.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start to 7 minutes	Retrieve the prerequisite and explain the purpose in the opening section.
7 to 20 minutes	Read the worked example and diagram; ask learners to trace each step aloud.
20 to 33 minutes	Complete W1 and W2, then compare predictions with observed results.
33 to 45 minutes	Collect the guided prediction before running; discuss a mismatch and its cause.
Session 2 start to 8 minutes	Retrieve the main rule without showing the previous answer.
8 to 20 minutes	Use W3 to diagnose and repair the stated failure; rerun an earlier case.
20 to 35 minutes	Collect W4 independently, including code or plan, predictions and explanations, before hints.
35 to 45 minutes	Discuss W5 and the exit; use the fresh retry when support was needed.

Reduce help deliberately

Completion practice

Completion task: Classify {"minutes": "5"} and {minutes: 5} as valid **JSON** text or malformed **JSON**.

Reasoning to listen for

Teacher check: The first is valid **JSON** but its minutes field fails the integer schema. The second is malformed because the key is not double-quoted.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a **response** from every learner before discussion.

Individual reasoning check

Use the worked case for **JSON** and **interface contract**s. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected repair

Expected correction: An **interface** should state what happens for malformed text, wrong structure and invalid values. Do not expose raw private **input** in error messages. Use only the fictional records supplied for practice.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Would converting the text silently prove the source count is correct? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Reasoned teacher answer

No. Under the stated strict **contract** reject the wrong type. Any explicit conversion policy would change the **contract**, and factual verification remains separate.

Changed-case check: Reject the **response** under this strict **contract**. Digits inside quotes represent text, whereas portions requires an integer.

Changed-case reasoning: Reject the **response** under this strict **contract**. Digits inside quotes represent text, whereas portions requires an integer.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

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Run the weekly evidence lab

> Run the Requests Responses and JSON investigation and use the first unsupported step to choose a teaching response.

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.

Inspect eight objects for syntax, required fields, types, allowed values, and source ownership; repair without inventing data.

> What did the hardest test teach you that the normal test did not?

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing.

Keep the expected result separate from an observed mistake.

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose and reteach

For guessing, return to given facts and **request** a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

A successful run proves the conclusion: An **interface** should state what happens for malformed text, wrong structure and invalid values. Do not expose raw private **input** in error messages.

Use only the fictional records supplied for practice.

Matching the example is enough: Require the new W4 case, explicit **prediction**s and an explanation. Fresh retry: label Tray and score integer 0 through 5.

Cases score 0 accepted, score 6 rejected, score null rejected; **JSON** list rejected before field access.

A convenient shortcut preserves meaning: Single quotes around property names are not valid **JSON** syntax. Use double quotes in the **JSON** text.

If the whole parsed value is a list, do not call .get on it: check the top-level type first and report a schema failure.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

Assess and explain the answers

> Use the Requests Responses and JSON questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

Worked case answer

Teacher key · Worked case answer

Request success, **JSON** validity, matching id, required fields and numeric type. Handle malformed **response**s explicitly.

Independent question

Independent question · withhold the key

An API returns {"ok":false,"error":"not_found"}. Your app tries to display **response**.title. What should it do instead?

Expected independent answer

Teacher key · Expected independent answer

Check the success or error fields before using result data. Show a useful not-found **response** and allow correction of the **request**.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “Every API **response** contains successful result fields.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Teacher key · Original transfer answer

The **response** is structurally plausible but for the wrong ID. Check **status**, parseability, required fields, types and matching ID before use.

Reject or report the mismatch; do not silently attach another book's page count to id=7.

Explanations for the additional workbook examples

W1 Translate a value

Workbook question · W1

What Python value does **JSON** null become? What **JSON** value represents Python True?

Teacher reasoning · W1

None and true, respectively.

Teacher reasoning · W1

Parsing JSON null produces Python None, while serialising Python True produces **JSON** true. The spelling belongs to the representation: lowercase true is **JSON** text, not the Python Boolean name True.

Keep **parsing** and serialisation directions clear.

W2 Separate failures

Workbook question · W2

Classify {"minutes": "5"} and {minutes: 5} as valid **JSON** text or malformed **JSON**.

Teacher reasoning · W2

The first is valid **JSON** but its minutes field fails the integer schema. The second is malformed because the key is not double-quoted.

W3 Repair the interface

Workbook question · W3

A parser accepts a **JSON** list then crashes on .get. What check belongs before field access?

Teacher reasoning · W3

Require type(record) is dict and handle failure before any .get or keyed field access.

W4 Specify independently

Workbook question · W4

Define a **JSON contract** for {"label": "Box", "count": 4}: nonempty text label, integer count 0 through 10. Give a valid case, a wrong-type case and a missing-field case with expected acceptance.

Teacher reasoning · W4

Valid Box/4 accepted; count "4" rejected for type; absent count rejected as missing. State that malformed **JSON** and non-object top levels are also rejected.

W5 Limit parsing

Workbook question · W5

A **JSON** record parses successfully. Does that prove its **observation** is correct?

Teacher reasoning · W5

No. Syntax, schema validity and **observation**al truth are separate checks.

Teacher reasoning · W5

No. **Parsing** checks **JSON** syntax. The schema separately checks the expected fields, types and ranges.

Collection **evidence** is a third question: even a well-formed, schema-valid duration could have been measured or recorded incorrectly.

Assess independent transfer

Independent question · withhold the key

Selected case: Define a **JSON contract** for {"label": "Box", "count": 4}: nonempty text label, integer count 0 through 10. Give a valid case, a wrong-type case and a missing-field case with expected acceptance.

Selected case answer

> **Teacher key · Selected case answer**
>
> Valid Box/4 accepted; count "4" rejected for type; absent count rejected as missing. State that malformed **JSON** and non-object top levels are also rejected.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

> **Teacher key · Selected case answer**
>
> Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.
>
> Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

Connect and extend learning

> Connect Requests Responses and JSON to Verified Study Buddy, adapt support and plan a later changed-case check.

Verified Study Buddy

Use a stable **request/response contract** between **interface**, retriever, model, and verifier.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Support and stretch

Provide one short approved passage; separate instruction, **input**, **output** and **evidence**. Stretch: introduce an unsupported claim or conflicting source and validate claim by claim with a fallback.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 23: Block-to-Python Bridge. Retrieve today's **evidence** and identify what carries forward and what is new.

Support and fresh reassessment

Color **JSON** punctuation separately from values. Extension: serialize a valid result, parse it again and check the data values are preserved.

Independent W4 **evidence**: Valid Box/4 accepted; count "4" rejected for type; absent count rejected as missing. State that malformed **JSON** and non-object top levels are also rejected.

Record whether the learner needed no help, a hint or explanation. Fresh retry: label Tray and score integer 0 through 5. Cases score 0 accepted, score 6 rejected, score null rejected; **JSON** list rejected before field access.

Next week: Exceptions and recovery. Reuse this week's explicit **contract**s and **evidence**.

Plan the delayed return

Suggested return points in this level: ai-3/23, ai-3/25, ai-3/28. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent **response** and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Prepare, teach and assess

> Use this preparation summary alongside the detailed Requests Responses and JSON lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

Teaching target: Parse **JSON** into Python records

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual **response**s. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
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>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

Before class, solve the supported practice: Classify {"minutes": "5"} and {minutes: 5} as valid **JSON** text or malformed **JSON**.

Supported practice answer

Reasoned practice answer: The first is valid **JSON** but its minutes field fails the integer schema. The second is malformed because the key is not double-quoted.

Check every learner before discussion: Use the worked case for **JSON** and **interface contract**s. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: An **interface** should state what happens for malformed text, wrong structure and invalid values. Do not expose raw private **input** in error messages. Use only the fictional records supplied for practice.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extend the reasoning

Extension: Replace 12 with the text "12". Require a **prediction** and a reason before comparing. Expected result: Reject the **response** under this strict **contract**.

Reason: Digits inside quotes represent text, whereas portions requires an integer.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

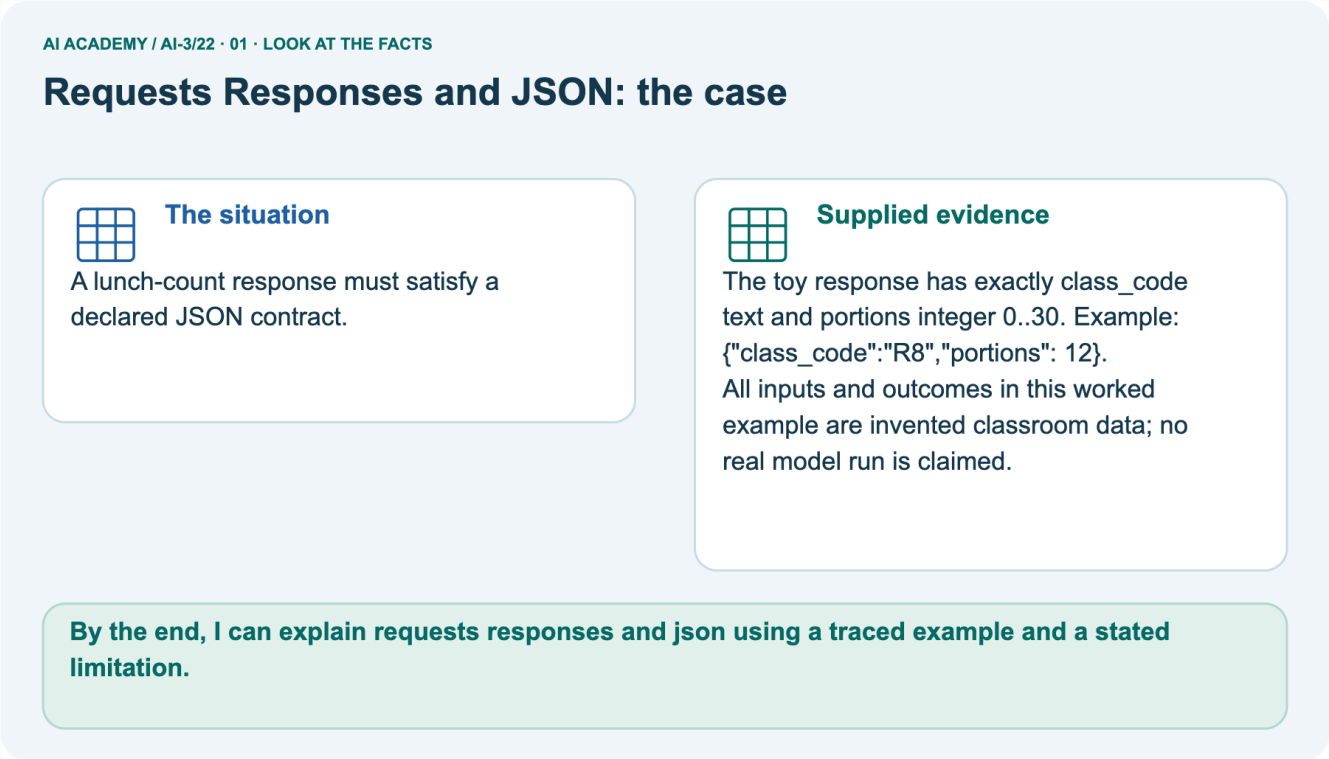
Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

Visual concept journey

> Show one Requests Responses and JSON graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Requests Responses and JSON: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

A lunch-count **response** must satisfy a declared **JSON contract**. Supplied facts: The toy **response** has exactly class_code text and portions integer 0..30. Example: {"class_code":"R8","portions": 12}.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

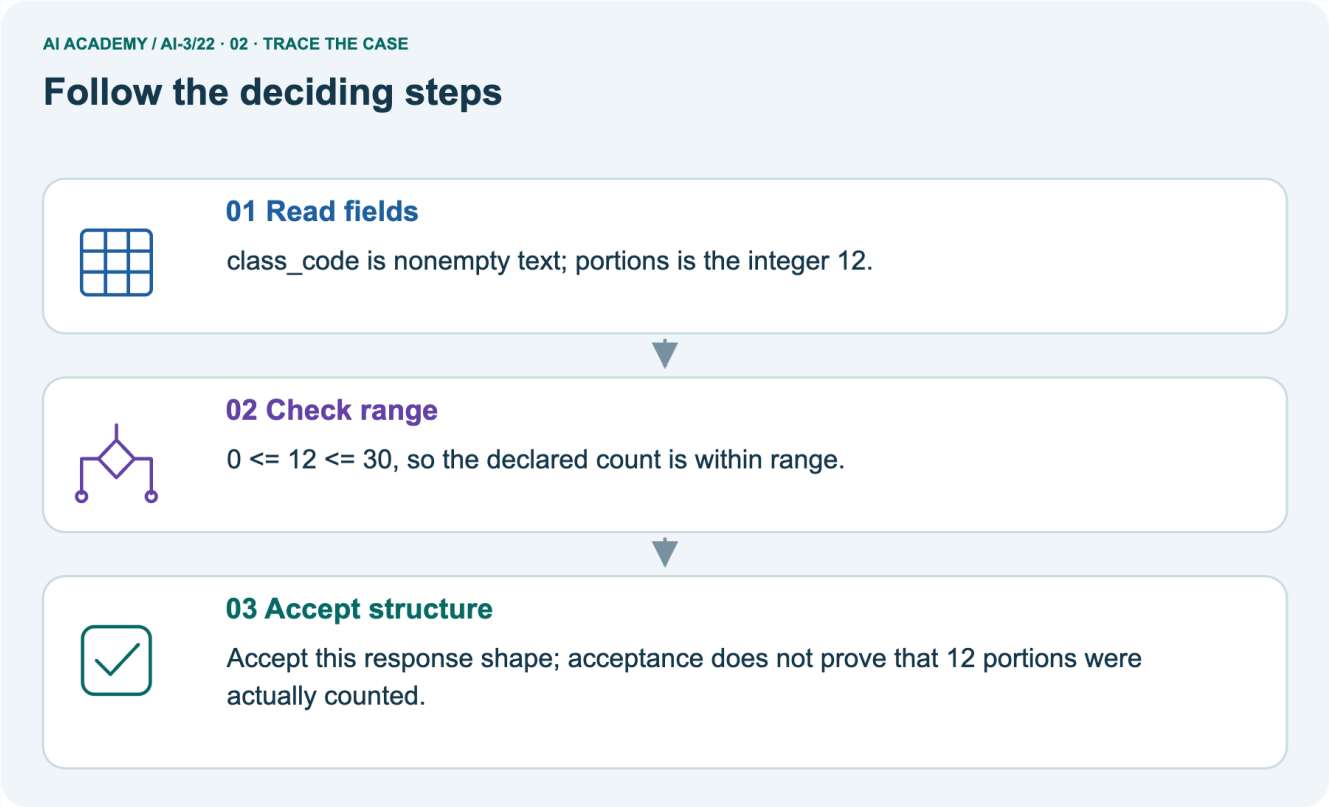
Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the

image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

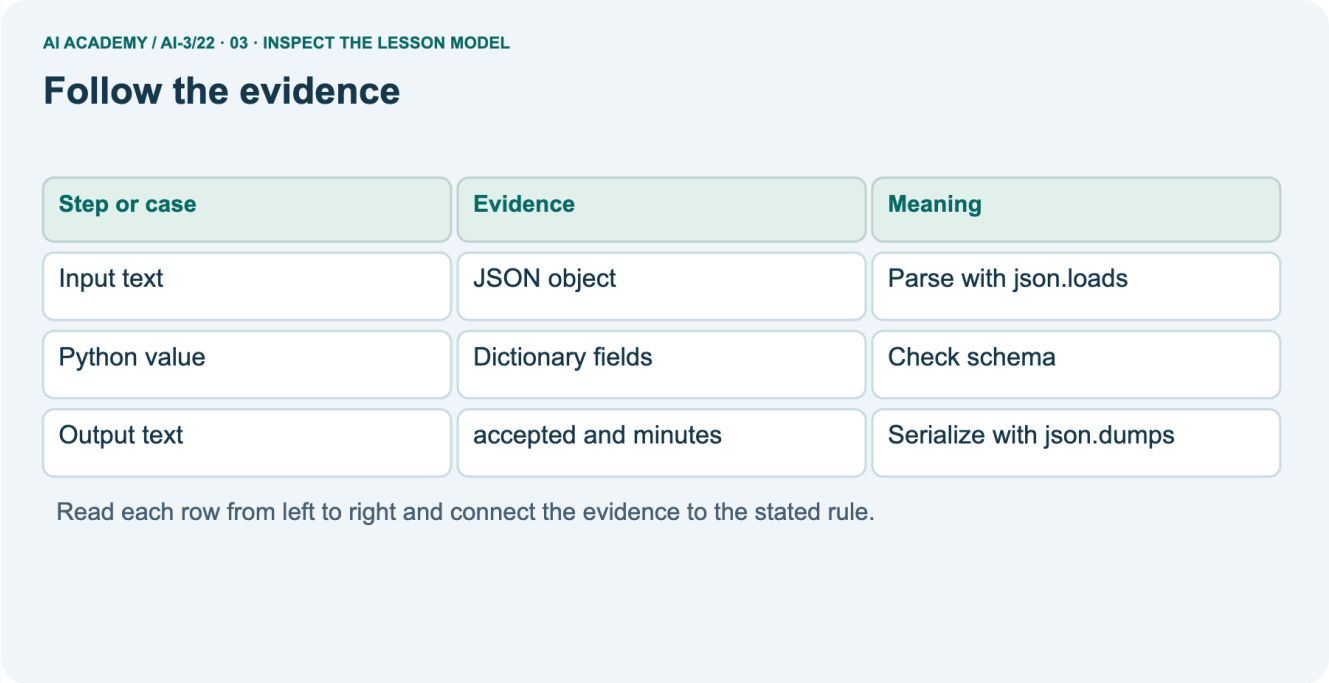
Read fields: class_code is nonempty text; portions is the integer 12. → Check range: 0 <= 12 <= 30, so the declared count is within range. → Accept structure: Accept this **response** shape; acceptance does not prove that 12 portions were actually counted.

Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Follow the evidence



Follow the labelled picture with the matching explanation. Open the vector diagram

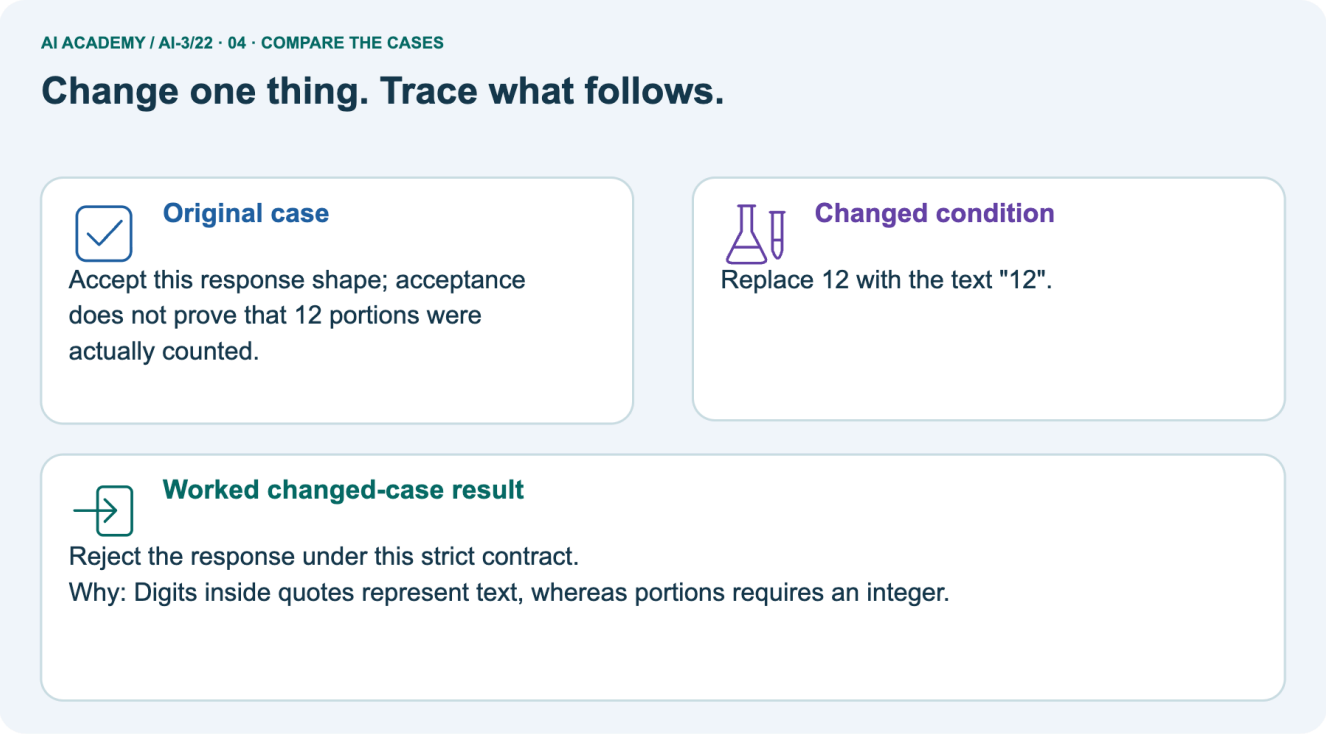
Follow the **evidence**. **Input** text | **JSON object** | Parse with **json**.loads. Python value | Dictionary fields | Check schema. **Output** text | accepted and minutes | Serialize with **json**.dumps

Read the labels and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Accept this **response** shape; acceptance does not prove that 12 portions were actually counted. Changed condition: Replace 12 with the text "12". Expected result: Reject the **response** under this strict **contract**.

Reason: Digits inside quotes represent text, whereas portions requires an integer.

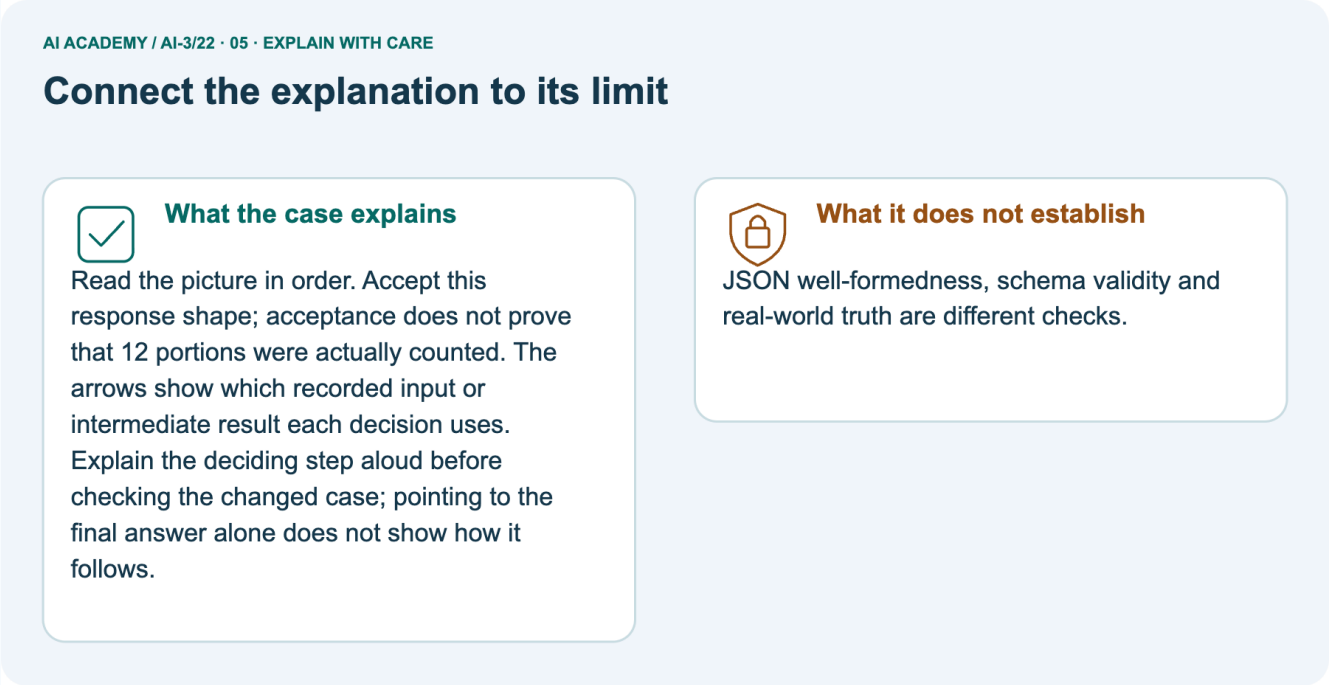
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Digits inside quotes represent text, whereas portions requires an integer.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Accept this **response** shape; acceptance does not prove that 12 portions were actually counted. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows. Limit: **JSON** well-formedness, schema validity and real-world truth are different checks.

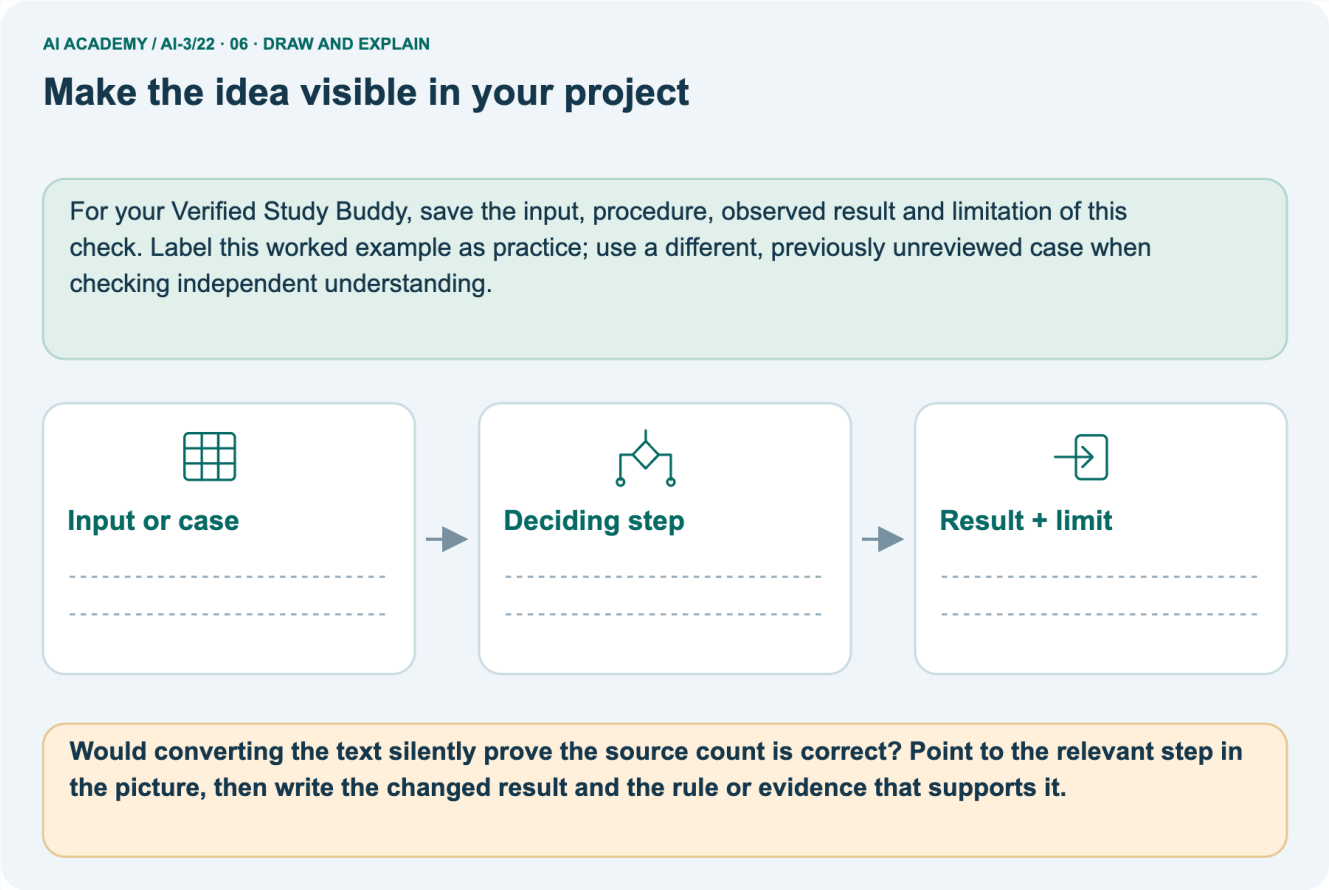
Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: **JSON** well-formedness, schema validity and real-world truth are different checks.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Would converting the text silently prove the source count is correct? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: No. Under the stated strict **contract** reject the wrong type. Any explicit conversion policy would change the **contract**, and factual verification remains separate.

Changed-case check: Reject the **response** under this strict **contract**. Digits inside quotes represent text, whereas portions requires an integer.